

7

Regression and Correlation Analysis

7
(2)

*Introduction to
Correlation Analysis*

7.2 Introduction to Correlation Analysis

In regression analysis, we shall develop an estimating equation – that is, a mathematical formula that relates the dependent variable to the independent variable.

7.1 *Simple Linear Regression Analysis*

Then, after we have learned the pattern of this relationship, we can apply *correlation analysis* to determine the strength of that relationship.

i.e. how well the estimating equation actually describes the relationship.

7.2 Introduction to Correlation Analysis

- Two variables are said to be correlated if the change in one variable results in a corresponding change in the other variable..
- The correlation is a statistical tool which studies the relationship between two variables.

7.2 Introduction to Correlation Analysis

- Correlation analysis involves various methods and techniques used for studying and measuring the extent of the relationship between the two variables.
- Correlation is concerned with the measurement of “strength of association between two variables.

7.2 Introduction to Correlation Analysis

- The degree of association between two or more variables is termed as ***correlation***.
- ***Correlation analysis*** helps us to decide the strength of the linear relationships between two variables.
- The word ***correlation*** is used to decide the degree of association between two variables.

7.2 Introduction to Correlation Analysis

Thus, correlation is a statistical tool, with the help of which, we can determine whether or not two or more variables are correlated and if they are correlated, what is the degree and direction of correlation.

7.2 Introduction to Correlation Analysis

Definition 1:- Correlation is the measure of the extent and the direction of the relationship between two or more variables.

Example 1:- (a) Weight and height of children
(b) An increase the price of commodity by a decrease in the quantity demanded.

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Types of correlations

7.2 Introduction to Correlation Analysis

Types of correlation:

- ❑ Positive and negative correlation
- ❑ Simple, partial and multiple correlation
- ❑ Linear and non-linear correlation

7.2 Introduction to Correlation Analysis

(i) Positive and negative correlation:

- If both the variables are varying in the same direction i.e. if one variable is increasing and the other on average is also increasing or if one variable is decreasing, the other on average is also decreasing, correlation is said to be *positive*.

7.2 Introduction to Correlation Analysis

On the other hand, if one of the variable is decreasing, while the other is increasing or vice versa, correlation is said to be *negative*.

Example 2:- Positive Correlation

- ☐ Heights and weights
- ☐ Amount of rainfall and yield of crops

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- ❑ Price and supply of commodity
- ❑ Income and expenditure of luxury goods
- ❑ Blood pressure and age

Example 3:- Negative Correlation

- ❑ Price and demand of commodity
- ❑ The time students watch TV and the test score

7.2 Introduction to Correlation Analysis

- ❑ Physical exercise and weight loss.
- ❑ School achievement and days of absent.
- ❑ The older a man gets, the less hair that he has.



**The older a man gets, the
less hair that he has.**

7.4

Linear and Non-Linear correlations

7.3 Types of correlations

Linear and Non Linear correlation

□ If the change in one variable tends to bear a constant ratio to the change in the other variable, the correlation is said to be *linear*.

Example:- The more time you spend running on a treadmill, the more calories you will burn.

7.3 Types of correlations

- ❑ Taller people have larger shoe sizes and shorter people have smaller shoe sizes.
- ❑ The longer your hair grows, the more shampoo you will need.
- ❑ The time students watch TV and the test score

7.3 Types of correlations

Correlation is said to be *non-linear* if the amount of change in one variable does not bear a constant ratio to the amount of change in the other variable.

7.3 Types of correlations

Example 1. Study Time and Exam Score (Diminishing Returns). A student increases daily study hours before an exam. From 1 to 4 hours: exam scores increase rapidly
Beyond 6 hours: improvement becomes very small due to fatigue

- Conclusion: Scores increase at first, then level off → non-linear correlation

7.3 Types of correlations

Example 2. Fertilizer Use and Crop Yield.

A farmer applies fertilizer to wheat crops. Small amounts increase yield significantly but Excess fertilizer damages crops and reduces yield.

- Conclusion: Yield first rises, then falls → non-linear (inverted U-shaped) correlation

7.3 Types of correlations

Simple, Partial and Multiple correlation:

- When only two variable are studied, it is the case of *simple correlation*.
- In the case of *partial and multiple correlation*, three or more variables are studied.

7.3 Types of correlations

Example 4:- you can use partial correlation to understand whether there is a linear relationship between ice cream sales and price, whilst controlling for daily temperature (i.e., the continuous dependent variable would be "ice cream sales", measured in US dollars, the continuous independent variable would be "price", also measured in US dollars, and the single control variable – that is, the single continuous independent variable you are adjusting for – would be daily temperature, measured in °C).

7.3 Types of correlations

You may believe that there is a relationship between ice cream sales and prices (i.e., sales go down as price goes up), but you would like to know if this relationship is affected by daily temperature (e.g., if the relationship changes when taking into account daily temperature since you suspect customers are more willing to buy ice creams, irrespective of price, when it is a really nice, hot day).

7.3 *Types of correlations*

In this course we are
going to dicuss simple
linear correlation.

7.4

Methods of studying correlation

7.4 Methods of studying correlation

The following are the methods of determining correlation:

1. Scatter diagram method

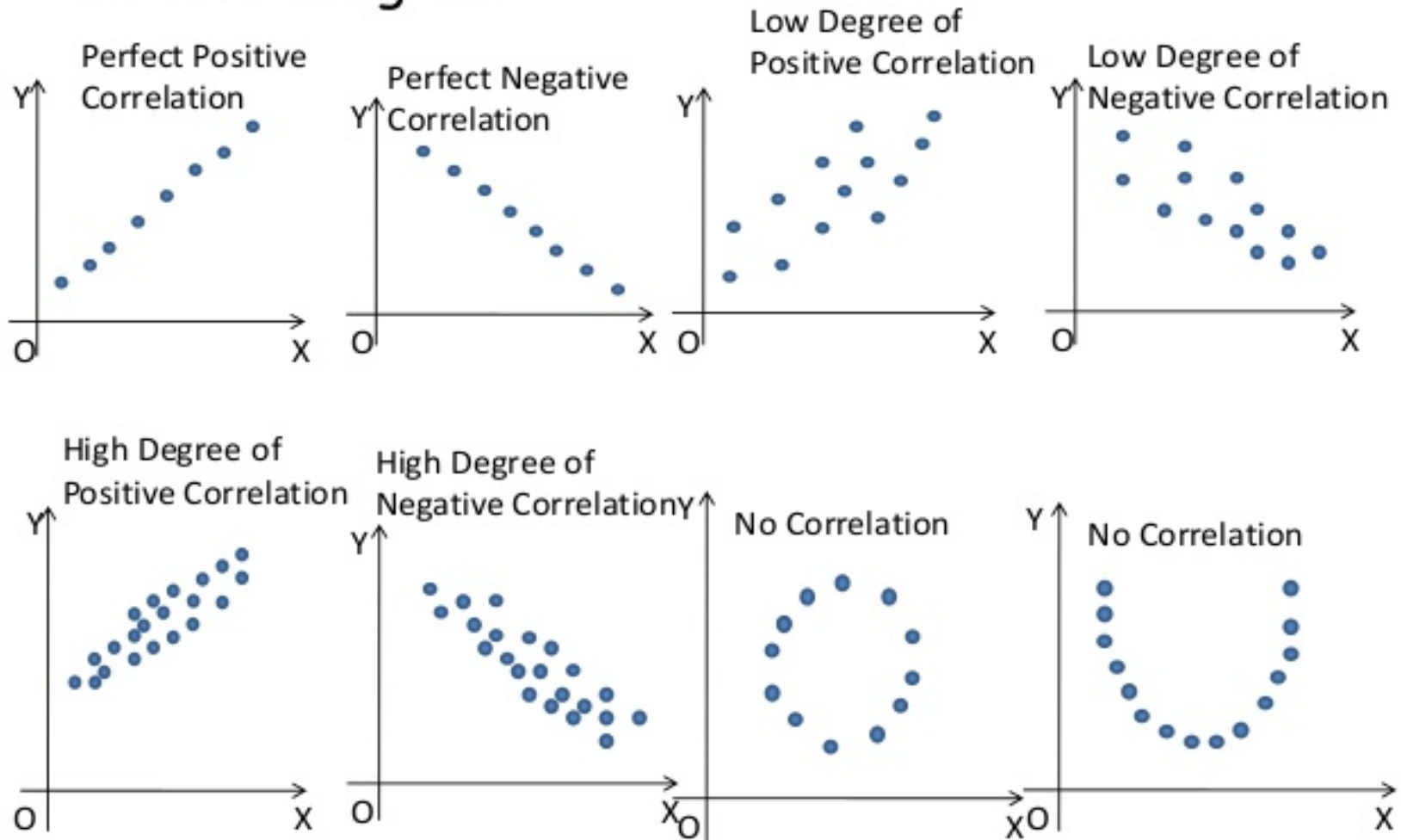
This is a graphic method of finding out relationship between variables.

7.4 Methods of studying correlation

To find a scatter diagram, given data are plotted on a graph paper in the form of dots i.e. for each pair of x and y values we put a dot and thus obtain as many points as the number of observations. *The greater the scatter of points over the graph, the lesser the relationship between the variables.*

7.4 Methods of studying correlation

Scatter Diagram



7.4 Methods of studying correlation

Interpretation of a scatter diagram

- If all the points lie on a straight line, there is either *perfect positive* or *perfect negative* correlation.
- If all the points lie on a straight falling from the lower left hand corner to the upper right hand corner, then the correlation is *perfect positive*.
- Perfect positive if $r = 1$.

7.4 Methods of studying correlation

- If all the points lie on a straight falling from the upper left hand corner to the lower right hand corner, then the correlation is *perfect negative*.
- Perfect negative if $r = -1$.
- The nearer the points are to the straight line, the higher the degree of correlation.

7.4 Methods of studying correlation

- The *farther* the points are from the straight line, *the lower the degree of correlation.*
- If the points are widely scattered and no trend is revealed, the variables may be un-correlated.
i.e. $r \approx 0$.

7.4 Methods of studying correlation

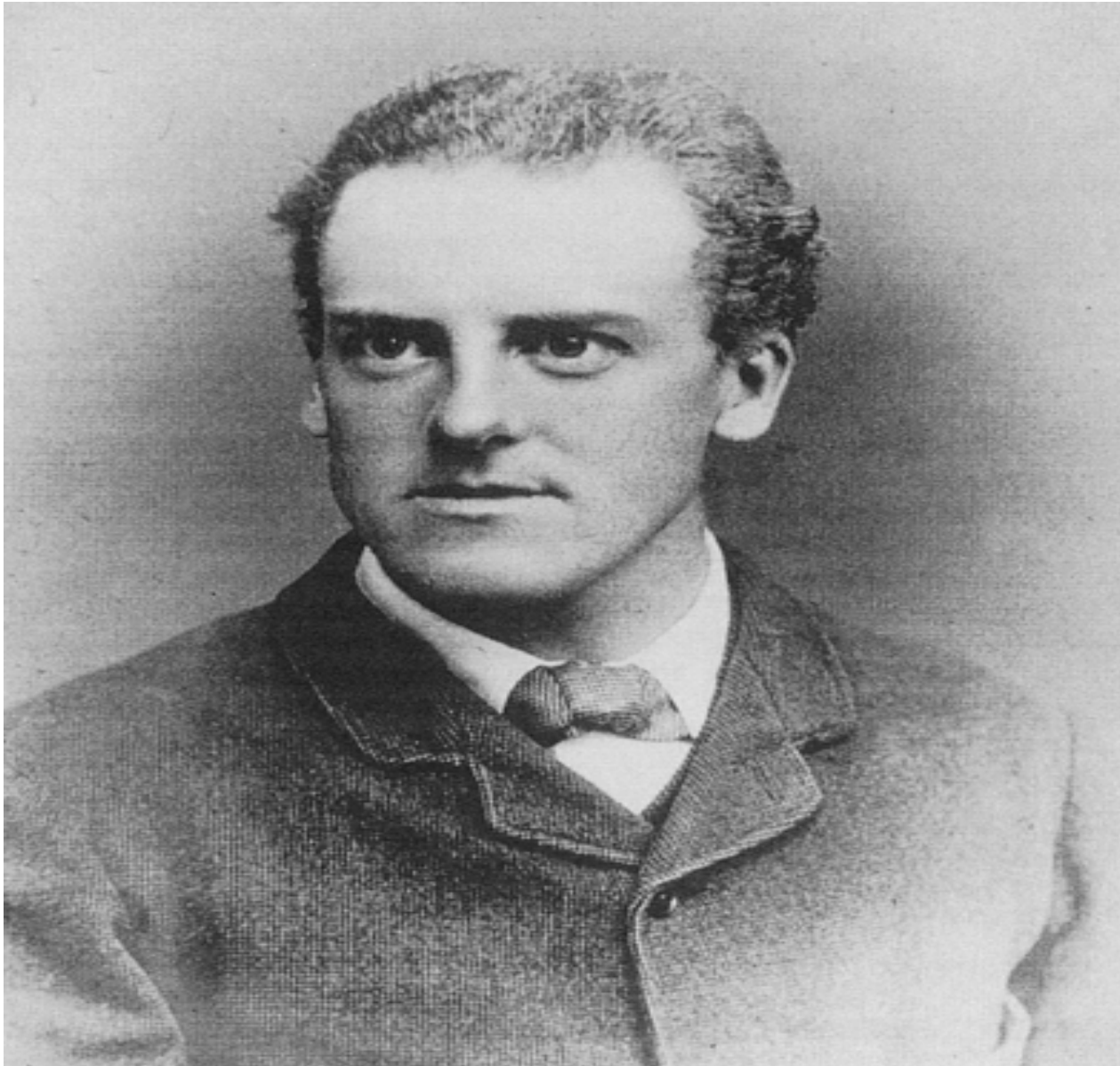
2. Karl Pearson's coefficient of correlation.

- A scatter diagram gives an idea about the type of relationship or association between the variables under study. It does not tell us about the quantification of the association between the two.

7.4 *Methods of studying correlation*

- In order to quantify the relationship between the variables a measure called *a correlation coefficient* was developed by Karl Pearson (1857 – 1936).

7.4 Methods of studying correlation



7.4 Methods of studying correlation

- Thus, the coefficient of correlation is a number which indicates to what extent two variables are related to what extent variations in one go with the variation of the other.
- The formula for simple correlation coefficient is calculated by the following formula:

$$r = \frac{N\sum xy - \sum x \sum y}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$$

7.4 Methods of studying correlation

Properties of Karl Pearson's correlation coefficient

1. The coefficient of correlation “ r ” is always a number between 1 and -1 inclusive.
2. If $r = 1$ or $r = -1$, the sample points lie on a straight line.

7.4 Methods of studying correlation

3. If r is nearer to 1 or -1 , there is a strong linear association between the variables.
4. If r is small (close to zero), there is low degree of correlation between the variables.

7.4 Methods of studying correlation

Interpretation of correlation coefficient

1. The coefficient of correlation, as obtained by the above formula shall always lie between -1 and 1 inclusive. i.e. $-1 \leq r \leq 1$.
2. When $r = +1$, there is a perfect positive correlation between the variables.

7.4 Methods of studying correlation

3. When $r = -1$, there is a perfect negative correlation between the variables.
4. When $r = 0$, there is no correlation between the variables.
5. When $0.7 \leq r \leq 0.99$, there is a high degree of correlation between the variables (in fact when $0.9 \leq r < 1$, we say that there is a very strong correlation between the variables).

7.4 Methods of studying correlation

6. When $0.5 \leq r \leq 0.699$, there is a moderate degree of correlation between the variables.
7. When $-0.5 < r < 0.5$, there is a low degree of correlation between the variables.
8. The correlation coefficient is a pure number. It is independent of the unit of measurement.

7.4 Methods of studying correlation

Now let's return to Example 2 of Chapter VII₍₂₎
(slide 41 and 42).

Before estimating the cost of producing 7 items,
we first find the coefficient of correlation, r ,
between production and cost incurred them.

7.1 Simple Linear Regression Analysis

Example 2:- The following table shows the number of items produced and the cost incurred in producing them (in birr).

Number of items produced	Cost incurred (in birr)
4	15
5	18
6	18
8	20
9	22

7.1 Simple Linear Regression Analysis

- *Find the equation of the regression line*
- *Identify the slope and the y-intercept and interpret them.*
- **Estimate the cost of producing 7 items.**

Number of items to be produced (x)	Cost incurred (y)	xy	x^2
4	15	60	16
5	18	90	25
6	18	108	36
8	20	160	64
9	22	198	81
$\sum_1^5 x_i = 32$	$\sum_1^5 y_i = 93$	$\sum_1^5 x_i y_i = 616$	$\sum_1^5 x_i^2 = 222$

7.4 Methods of studying correlation

Number of items to be produced (x)	Cost incurred (y)	xy	x^2	y^2
4	15	60	16	225
5	18	90	25	324
6	18	108	36	324
8	20	160	64	400
9	22	198	81	484
$\sum_1^5 x_i = 32$	$\sum_1^5 y_i = 93$	$\sum_1^5 x_i y_i = 616$	$\sum_1^5 x_i^2 = 222$	$\sum_1^5 y_i^2 = 1757$

7.4 Methods of studying correlation

$$\begin{aligned} r &= \frac{N\sum xy - \sum x \sum y}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}} \\ &= \frac{5(616) - (32)(93)}{\sqrt{[5(222) - (32)^2][5(1757) - (93)^2]}} \\ &= \frac{104}{\sqrt{[86][136]}} = \frac{104}{\sqrt{11696}} = \frac{104}{108.15} = 0.96 \end{aligned}$$

7.4 Methods of studying correlation

Thus, we can say that there is a very strong positive relationship between the number of items to be produced and the cost incurred to produce them. So the estimated equation of the regression line $\hat{y} = 10.86 + 1.21x$ can be used for estimation purpose.

7.4 Methods of studying correlation

Hence the estimated cost of producing 7 items will be:

$$\begin{aligned}\hat{y} &= 10.86 + 1.21(7) \\ &= 10.86 + 8.47 \\ &= 19.33 \text{ br}\end{aligned}$$

7.1 Simple Linear Regression Analysis

Example 2:- The Quality Assurance and Student Service of HiLCoE wants to determine whether there is any type of relationship between the amount of alumni's contribution (in tens of birr) and the years the alumnus has been out of schools. The following table refers to the contribution of the sample of five students and the number of years they graduated from HiLCoE.

7.1 Simple Linear Regression Analysis

Number of years	Contribution (in tens of birr)			
1	50			
5	10			
7	6			
3	30			
6	8			

7.5

*The coefficient of
determination*

7.4 Methods of studying correlation

The Coefficient of Determination

The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable. The coefficient of determination is also called “ R – squared” and is denoted by R^2 .

- Note that $0 \leq R^2 \leq 1$.

7.4 Methods of studying correlation

- The coefficient of determination, R^2 (or sometimes r^2) is another measure of how well the least squares equation $\hat{y} = b_0 + b_1x$ performs as a predictor of y .
- The higher the R^2 , the more useful the model.
 R^2 takes on values between 0 and 1.

7.4 *Methods of studying correlation*

- Essentially, R^2 tells us how much better we can do in predicting y by using the model and computing $\hat{y}$ than by just using the mean $\bar{y}$ as a predictor.
- Note that when we use the model and compute $\hat{y}$ the prediction depends on x because $\hat{y} = b_0 + b_1x$.
 - Thus, we act as if x contains information about y .

7.4 Methods of studying correlation

Explained and unexplained variations

- The total variation in the dependent variable (y) can be divided into two:
 - explained variation
 - unexplained variation

7.4 Methods of studying correlation

Explained variation:- is a change in the dependent variable (y) explained by changes in the independent variable (x). The proportion of explained variation is:

$$r^2 \times 100\%$$

7.4 Methods of studying correlation

Unexplained variation:- is the variation in the dependent variable (y) due to chance, excluded variables, etc. The proportion of unexplained variation is:

$$(1 - r^2) \times 100\%$$

7.4 Methods of studying correlation

Example 1:- The following data is the monthly amount of money spent on advertising (x) (in thousands of birr) and the monthly number of passengers (y) (in thousands), of a certain private airlines in Rwanda in randomly selected five months.

7.4 Methods of studying correlation

Advertising expense (x)	Number of passangers (y)
10	15
12	17
8	13
17	23
10	17

7.4 Methods of studying correlation

- i) Find the equation of the estimated regression line.
- ii) Identify the slope (gradient and interpret your result.
- iii) Estimate the number of passengers if the airline expends 15,000 birr for advertisement
- iv) Calculate the coefficient of determination
- v) Find the proportion of explained and unexplained variation and interpret.

7.4 Methods of studying correlation

Advertising expense (x)	Number of passengers (y)	x^2	y^2	xy
10	15	100	225	250
12	17	144	189	204
8	13	64	169	104
17	23	289	529	391
10	17	100	189	170
$\Sigma x = 57$	$\Sigma y = 85$	$\Sigma x^2 = 697$	$\Sigma y^2 = 1501$	$\Sigma xy = 1019$

7.4 Methods of studying correlation

Solution:- (i) The coefficient of correlation is:

$$\begin{aligned} r &= \frac{N\sum xy - \sum x \sum y}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}} \\ &= \frac{5(1019) - (57)(85)}{\sqrt{[5(697) - (57)^2][5(1501) - (85)^2]}} \\ &= \frac{250}{\sqrt{[236][280]}} = \frac{250}{\sqrt{66080}} = \frac{250}{257.1} = 0.9723 \end{aligned}$$

7.4 Methods of studying correlation

Hence the coefficient of determination is:

$$r^2 = (0.9723)^2 = 0.9458$$

This figure indicates that there is a very strong correlation between advertising expense and number of passengers.

7.4 Methods of studying correlation

ii) *a.* The proportion of explained variation is
 $r^2 \times 100\% = (0.9723)^2 \times 100\% = 94.58 \%$.

Thus, we can conclude that 94.58 % of the change in the number of passengers is explained by changes in amount of money spent on advertising.

7.4 Methods of studying correlation

b. The proportion of unexplained variation is

$$(1 - r^2) \times 100\% = (1 -$$

$$0.9458) \times 100\%$$

$$= 0.0542 \times 100\% = 5.42 \%$$

Thus, 5.42 % of the change in the number of

7.4 Methods of studying correlation

Example 2:- The table below gives the amount of time a sample of 6 students in a class studied for midterm exam and their results.

Hours studied	Results (Out of 40%)			
1	16			
2	24			
4	31			
6	32			
7	34			
8	38			

7.4 Methods of studying correlation

- i. Find the equation of the estimated regression line.
- ii. Identify the slope (gradient and interpret your result.
- iii. Estimate the mark of a student who studied for the exam for 3 hours.*
- iv. Find the explained variation and interpret your result.
- v. Find the unexplained variation and interpret your result.

7.4 Methods of studying correlation

Example 3: The following table refers to the number of hours six students watched foot ball on television during the weekends and the scores of each student who sat for the final exam in *Statistics and Probability* on the following Monday.

7.4 Methods of studying correlation

Hours spent	Test Score (Out of 100%)
4	77
2	83
4	70
3	75
6	60
1	90

7.4 Methods of studying correlation

- (i) Find the equation of the estimated regression line.
- (ii) Identify the slope (gradient and interpret your result.
- (iii) Estimate the mark of a student who watched TV during the weekends for five hours.
- (iv) Find the explained and unexplained variation and interpret each of your results.

7.4 Methods of studying correlation

Example 4 (Winter 2025 Final Examination): A

school administrator wants to understand how teacher experience affects student performance.

The following table refers to the teaching experience (in years) of five teachers and the average test marks of students (out of 15%).

7.4 Methods of studying correlation

Teaching experience (in years)	Test marks (out of 15%)		
2	6		
6	7		
8	9		
10	12		
14	13		

7.4 Methods of studying correlation

1. Find the estimated equation of regression line.
2. Find the slope (gradient) and the y-intercept and interpret each of them.
3. Estimate the average mark of students taught by a teacher with a teaching experience of 9 years.
4. Find the explained and unexplained variation and interpret.

Any Questions



Any Questions

